// STRATEGY RESEARCH NOTE

TWIN CONTINUATION

One trend-continuation system on NQ futures with two entry setups and one position at a time. A swing breakout and a morning momentum entry, each following logic documented in published market research.

INSTRUMENT	SAMPLE	PERIOD	TIMEFRAME
NQ Futures	**816 trades**	**2020 / 2026**	**1 Minute**
DIRECTION	ORIGIN	PLATFORM	SETUPS
Long & Short	**Research Papers**	**MultiCharts**	**2, one position**

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// ABSTRACT

THE CASE.

Twin Continuation is an intraday system on NQ futures built around one idea: **once a move is underway, it tends to continue**. The system has two entry setups, trades long and short, and holds one position at a time.

The **Swing Breakout** places stop orders at the latest untested swing high and swing low on 6-hour candles. Its logic follows Osler (2003): stop orders cluster just beyond support and resistance, and triggering them accelerates the break.

The **Morning Momentum** setup enters in the direction of a strong volatility-normalised extension in the first hour of the cash session and holds into the afternoon. Its logic follows Baltussen, Da, Lammers and Martens (2021): hedging flows trade in the direction of the move and make the day's direction persist.

Across **816 trades** from January 2020 to September 2026 the MultiCharts run shows **$334,239** net profit on one NQ contract, a profit factor of **1.73**, a 69.0% hit rate and a **$22,540** maximum strategy drawdown, with commission and slippage included. The out-of-sample period (2023 onward) is stronger than the in-sample period.

Bootstrapped on the Apex 50K evaluation, the system passes **43.2%** of single attempts and **67.8%** within two.

Regime-Dependent

The edge is tied to the trend-following regime NQ has been in since 2020. Tested back to 2016, the system ran flat until then. A regime switch back to mean reversion would likely switch the edge off (§08).

01

// RESEARCH AND TRANSLATION

WHERE IT COMES FROM.

// Setup A · Swing Breakout

Osler, C. L. (2003). "Currency Orders and Exchange Rate Dynamics: An Explanation for the Predictive Success of Technical Analysis." *Journal of Finance* 58(5), 1791–1819. Free version: Federal Reserve Bank of New York Staff Report No. 125.

> *Trends gain momentum once predictable support and resistance levels are crossed.*
>
> OSLER, STAFF REPORT 125, P. 1 (ABSTRACT)

Osler studies the full stop-loss and take-profit order book of a large dealing bank. Take-profit orders tend to reverse moves, stop-loss orders tend to intensify them. Stop-buy orders cluster just above key levels and stop-sell orders just below them (pp. 5–6, §II.D pp. 19–21, Table I p. 37), so price accelerates once a level is broken. Practitioners are also shown to place their stops just beyond support and resistance (p. 22).

// Translation Into The System

A swing high or low on higher-timeframe candles is a natural support/resistance level, and exactly where the market parks its stops. The system places its own **entry as a stop order at that level**, so it gets filled at the moment the stop cluster fires, and trades in the direction of the resulting acceleration.

// Setup B · Morning Momentum

Baltussen, G., Da, Z., Lammers, S. and Martens, M. (2021). "Hedging Demand and Market Intraday Momentum." *Journal of Financial Economics* 142, 377–403.

> *Hedging short gamma exposure requires trading in the direction of price movements.*
>
> BALTUSSEN ET AL., JFE 142, P. 377 (ABSTRACT)

Across more than 60 futures from 1974 to 2020, the return of the day up to the last half hour predicts the direction into the close. Option market makers and leveraged ETFs must hedge in the direction of the move, which makes it persist. NQ is explicitly in the sample (Table 1, p. 383) and shows one of the strongest results of all markets (Table B1, p. 397). The effect is present in both 1974–1999 and 2000–2020 (Table 5, p. 386).

// Translation Into The System

If the day's direction tends to persist, a strong extension early in the session is worth joining. The setup measures how far price has stretched from its mean **relative to volatility**, enters when that stretch becomes extreme in the first hour, and holds into the afternoon instead of taking a quick target.

// WHAT THIS IS NOT

The papers describe market mechanisms, not these trading rules. Swing length, volatility thresholds and time windows are design choices made for this system. Reading the papers explains *why* the setups are built the way they are; it does not reproduce them.

02

// AS IMPLEMENTED

SPECIFICATION.

// Setup A · Swing Breakout

- 6-hour candles aligned to the 17:00 CT session open. A swing high is a candle whose high is at least the highs of the three candles either side of it; lows mirror this.
- Buy stop at the latest untested swing high, sell stop at the latest untested swing low. Re-armed every 15 minutes while flat. A level is used once.
- Stop at the opposite swing. Target at fill ± 2 × ATR(14) on 15-minute bars.
- No new orders after 15:15 CT, hard flat 15:45 CT. Trades the overnight session.
- Volatility filter: no swing trades in a session whose previous session range exceeded 2 × its 50-session average.

// Setup B · Morning Momentum

- Oscillator on 10-minute buckets: distance of close from EMA(21), divided by 3 × Wilder ATR(14), smoothed by EMA(3). Runs continuously across sessions.
- Oscillator crosses above +100 → long, below −100 → short, on a complete bucket closing 08:40–09:30 CT.
- Market entry, fixed 75-point stop, no target, flat 14:50 CT. One attempt per day.

// One-Position Rule

A setup only enters while flat. If one setup holds a trade, the other setup's signal is ignored. Every exit is tied to its own entry, so one setup never closes the other's position.

// Inputs As Tested

Swing target	2.0 × ATR
ATR length	14
Swing candle	360 min
Volatility filter	2.0 × avg range
Volatility lookback	50 sessions
Momentum pivot / ATR / smooth	21 / 14 / 3
Momentum threshold	±100
Momentum stop	75 points
Size	1 contract

FIG. A STRATEGY INPUTS

// Data & Test Environment

Symbol	NQ (continuous)
Chart	1 Minute, Data1 only
Session / time zone	24h Globex, CT
Start / end	2020-01-06 / 2026-09-29
Initial capital	$50,000
Commission / slippage	$5 / $1 per trade
Point value	$20
Intrabar order generation	Off

FIG. B ENVIRONMENT AS RECORDED IN THE WORKBOOK

03

// ANALYSING THE RESULT

DOES IT MAKE MONEY?

// NET PROFIT	// PROFIT FACTOR	// MAX STRATEGY DD	// ANN. SHARPE
$334,239	**1.73**	**$22,540**	**2.12**
816 trades, costs included	adj. 1.56 / select 1.78	intraday peak-to-trough, Mar 2022	monthly basis, MultiCharts

PERFORMANCE SUMMARY	ALL TRADES
Net Profit	**$334,239**
Gross Profit	$794,215
Gross Loss	($459,976)
Adjusted Net Profit	$271,848
Long Net Profit	$153,615
Short Net Profit	$180,624
Max Strategy Drawdown	($22,540)
Max Close-to-Close DD	($17,091)
Return on Max Strategy DD	**14.83**
Largest Winner	$11,133
Largest Loser	($11,712)

RATIOS AND RATES	ALL TRADES
Profit Factor	**1.73**
Percent Profitable	**68.99%**
Avg Trade	$409.61
Avg Winner / Avg Loser	$1,411 / ($1,818)
Avg Monthly Return	$4,126
Monthly Return StDev	$6,605
Sharpe / Sortino	0.61 / 1.22
Max Consecutive Winners / Losers	12 / 5
Positive Months	59 of 81 (72.8%)
Best / Worst Month	$20,340 / ($11,163)
Total # of Trades	**816**

// TIME IN MARKET

Trading period 6 years, 9 months. Position held only **4.5%** of elapsed time; the system is selective by design.

// COSTS PAID

Commission **$8,160**, slippage **$1,256**. Both are inside the net profit figure above.

// DIRECTION

444 long and 372 short trades. Shorts earn slightly more than longs, so the result is not a pure bet on a rising index.

04

// VISUAL INSPECTION

EQUITY / DRAWDOWN.

The curve climbs from $50k to roughly $384k on a fixed single contract. No calendar year lost money, and the ascent is steady across the in-sample and out-of-sample periods.



FIG. 1 EQUITY CURVE, $50K BASE, ONE NQ CONTRACT, JAN 2020 TO SEP 2026. DASHED LINE: 2023 IN-SAMPLE / OUT-OF-SAMPLE BOUNDARY

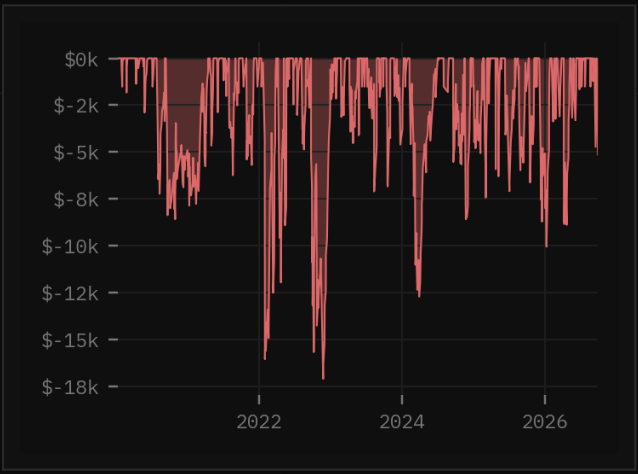


FIG. 2 UNDERWATER CURVE, CLOSED TRADES

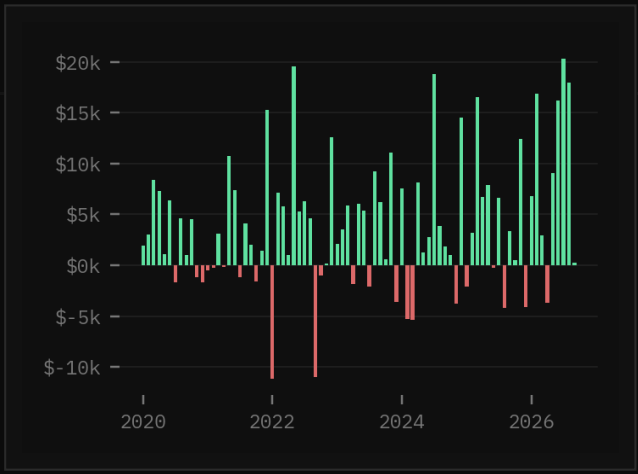


FIG. 3 NET PROFIT PER MONTH

// DRAWDOWN ENVELOPE

The deepest drawdown sits in **early 2022**, the bear-market start, at **$22,540** intraday and $17,091 closed. It sits in the in-sample period, not at the right-hand edge of the curve.

// MONTHLY PROFILE

59 of 81 months positive (72.8%). Best month **$20,340**, worst month **($11,163)**, average month $4,126.

05

// PER-TRADE GEOMETRY

DISTRIBUTION.

Across 816 trades the system wins **69.0%** of the time with an average winner of **$1,411** against an average loser of **$1,818**. This is a hit-rate system: winners are smaller than losers, and the edge comes from winning often. The median trade is **$679**.

P&L BUCKET	TRADES
Below −$5,000	15
−$5,000 to −$2,500	27
−$2,500 to −$1,000	150
−$1,000 to $0	61
$0 to $1,000	**277**
$1,000 to $2,500	219
$2,500 to $5,000	48
Above $5,000	19

DERIVED FROM THE EXPORTED TRADE LOG

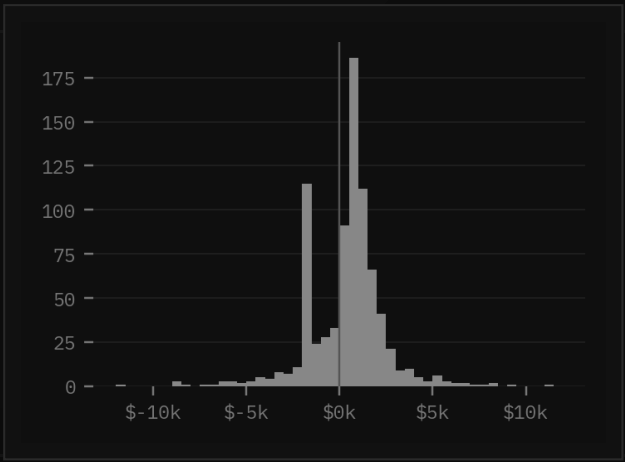


FIG. 4 PER-TRADE P&L DISTRIBUTION, $500 BINS

// CONCENTRATION

The largest winner, $11,133, is 3.3% of net profit; the five largest together are 13.3%. Removing them still leaves **$289,879**. No single trade carries the result.

EXIT	TRADES	NET P&L	AVG
Target	376	$420,054	$1,117
Timed flat	320	$153,415	$479
Stop loss	120	($239,230)	($1,994)

DIRECTION	TRADES	NET P&L	PF
Long	444	$153,615	1.69
Short	372	$180,624	1.76

Stops are rare. Only 120 of 816 trades, 14.7%, end on a stop loss. The rest close at a target or at the timed flat in the afternoon, and the timed flat is profitable on average: trades still open late in the session are usually running in the right direction.

Both directions earn. Shorts make slightly more than longs at a similar profit factor, so the result is not a disguised long bet on a rising index.

Short holding time. The median trade lasts 136 minutes; winners resolve faster (115 min) than losers (175 min).

06

// DEVELOPMENT PERIOD VS LATER PERIOD

SPLIT SAMPLE.

Development and parameter selection were carried out on **January 2020 to December 2022**. The following period, **January 2023 to September 2026**, was used afterwards to check whether the logic held outside the development window. The split is computed directly from the exported trade log.

METRIC	IN-SAMPLE	OUT-OF-SAMPLE
Period	2020-01 / 2022-12	2023-01 / 2026-09
Trades	**343**	**473**
Net profit	**$113,245**	**$220,994**
Profit factor	**1.56**	**1.85**
Win rate	**69.10%**	**68.92%**
Avg trade	$330	$467
Avg winner	$1,326	$1,473
Avg loser	($1,895)	($1,762)
Max closed-trade DD	($17,091)	($12,714)
Largest loser	($11,712)	($8,677)

DRAWDOWN IS CLOSED-TRADE, NOT INTRABAR

The out-of-sample period is the stronger one. Profit factor rises from 1.56 to 1.85, average trade from $330 to $467, and the drawdown shrinks. Win rate is practically unchanged at 69%, which is what a stable mechanism looks like.

Two qualifications. This is a single chronological split, not a walk-forward. And the later period coincides with the strongest trend-following years in NQ, so part of the improvement is regime, not skill (§08).

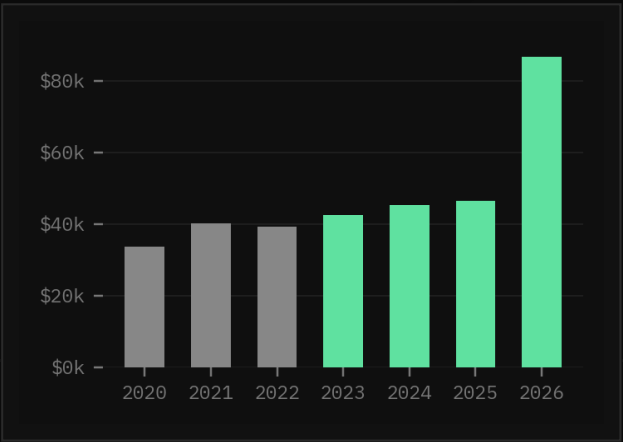


FIG. 5 NET PROFIT PER YEAR. GREY IN-SAMPLE, GREEN OUT-OF-SAMPLE

YEAR	TRADES	NET P&L	PF	WIN %
2020	103	$33,765	1.77	72.8
2021	120	$40,195	1.81	66.7
2022	120	$39,285	1.37	68.3
2023	137	$42,530	1.69	67.9
2024	123	$45,362	1.75	71.5
2025	126	$46,514	1.57	65.9
2026 (to Sep)	87	$86,588	2.56	71.3

NO LOSING CALENDAR YEAR; 59 OF 81 MONTHS POSITIVE (72.8%)

07

// 20,000 SIMULATED EVALUATIONS

PROP-FIRM BOOTSTRAP.

The trade log was resampled by trading day into **20,000 simulated Apex 50K evaluations**: $3,000 profit target, $2,000 end-of-day trailing drawdown, $1,000 daily loss limit, 30 calendar days. The drawdown floor is checked at each trade's worst intraday point, not only at the close, so losing trades that later recover still count against the account.

// PASS, 1 ATTEMPT	// WITHIN 2	// WITHIN 3	// MEDIAN DAYS
43.2%	**67.8%**	**81.7%**	**7**
95% CI 42.5% – 43.9%	independent attempts	independent attempts	weekdays to pass, avg 8.4

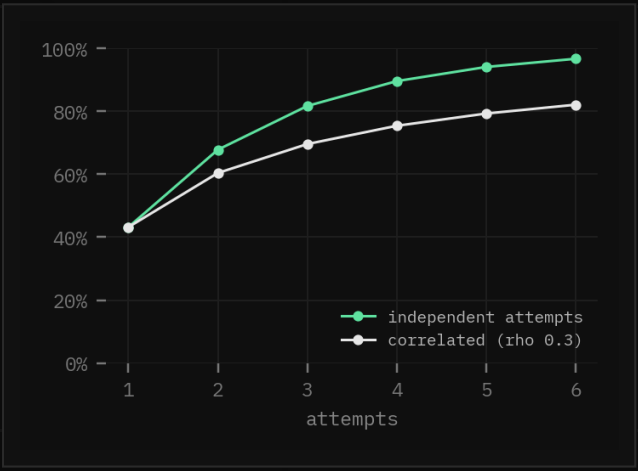


FIG. 6 PROBABILITY OF PASSING WITHIN N ATTEMPTS

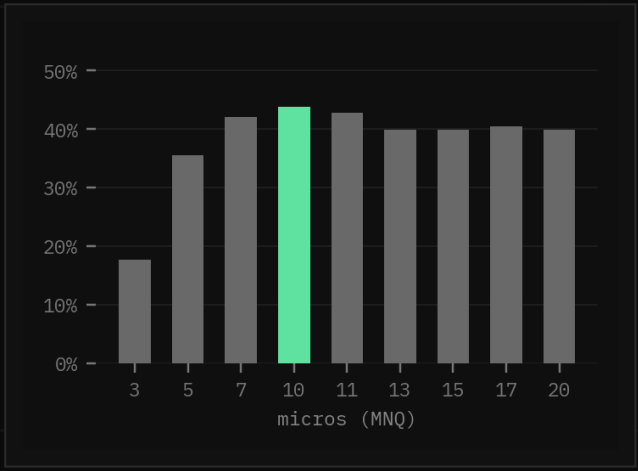


FIG. 7 SINGLE-ATTEMPT PASS RATE BY SIZE

ATTEMPTS	INDEPENDENT	CORRELATED
1	43.2%	43.2%
2	67.8%	60.4%
3	81.7%	69.6%
4	89.6%	75.4%
5	94.1%	79.3%

CORRELATED: ATTEMPTS SHARE A COMMON FATE, RHO 0.3

Size: 10 micros (MNQ), equivalent to one NQ. Smaller sizes pass less often because they run out of time; larger sizes stop adding pass rate and only add drawdown.

Failure mode: 51.7% of runs fail on the trailing drawdown, 5.0% on time. A passing run traded on 3.6 days on average.

Robustness: resampling in 5- and 10-day blocks, which keeps losing streaks intact, gives 44.5% and 43.5%, matching the day-by-day result.

08

// WHAT THIS DOCUMENT DOES NOT SHOW

REGIME DEPENDENCE.

// The System Needs A Trending Market

The system trades continuation. It earns when moves extend and lose when moves snap back. From 2020 onward NQ has behaved in a trend-following way, and that is the regime this sample covers.

The system was also run **back to 2016**. From 2016 to 2019 it went sideways: it did not lose, but it barely made anything, only around $20,000–$30,000 over four years on one contract. The edge was not there.

This matches the research. Baltussen, van Bekkum and Da (2019, *Journal of Financial Economics* 132, 26–48) document that short-term serial dependence in major stock indexes turned from positive to negative since the 2000s, linked to the rise of index products. A negatively autocorrelated, mean-reverting tape is exactly where continuation setups stall.

> *If NQ switches back to a mean-reverting regime, this system will most likely go flat again. It should be run with that expectation, not against it.*
>
> CONCLUSION OF §08

// Known Limitations

Evidential base	One MultiCharts export
Split test	Single split, no walk-forward
Parameter sensitivity	Only the volatility filter swept
Fills	1-minute bars, no tick data
Slippage	$1 per trade, below one tick
Limit orders	Filled on touch
Bootstrap	Resamples history, not new regimes

// Tail Risk

The largest single loss is **$11,712** on one NQ, a swing trade whose stop sits at the opposite swing and can be far away. The volatility filter removes the worst crash sessions, but not every wide stop. On a prop account the size has to be chosen with this tail in mind, not with the average trade.

// The Papers Are Not The Rules

Osler (2003) and Baltussen et al. (2021) support the mechanisms. They do not test these parameters, this instrument's 1-minute data or this combination.

09

// STATUS OF THE MODEL

VERDICT.

Validated / Regime-Bound

One continuation system with two research-grounded setups, 816 trades over almost seven years, no losing year and a stronger out-of-sample than in-sample period. It works in the regime it was built for and is not expected to work outside it.

// WHAT HOLDS UP

Profit factor 1.85 out-of-sample against 1.56 in-sample.

Profitable long and short, no losing calendar year.

43.2% Apex 50K pass rate, 67.8% within two attempts.

// WHAT TO WATCH

Regime: flat from 2016 to 2019.

Swing stops can be wide; largest loss $11,712 on one NQ.

Deepest drawdown $22,540, early 2022.

// WHAT IS MISSING

No walk-forward.

No tick-level fill reconstruction.

No test on other index futures.

// Build Something Of Your Own

This is not a finished product to copy and trade. The logic is published so it can be taken apart. Places to start:

- **The breakout level.** 6-hour swings are one choice. Prior-day high/low, weekly levels or round numbers, the levels Osler actually studied, are all open.
- **The momentum trigger.** Baltussen et al. use the return up to the last half hour. An entry later in the day, closer to their setup, is untested here.
- **The regime switch.** A filter that detects mean-reverting phases and stands the system down would address its main weakness.
- **The pass rate.** Sizing, daily stops or a third uncorrelated setup could push it higher.

// References

Osler, C. L. (2003). Currency Orders and Exchange Rate Dynamics: An Explanation for the Predictive Success of Technical Analysis. *Journal of Finance* 58(5), 1791–1819. FRBNY Staff Report No. 125.

Baltussen, G., Da, Z., Lammers, S., Martens, M. (2021). Hedging Demand and Market Intraday Momentum. *Journal of Financial Economics* 142, 377–403.

Baltussen, G., van Bekkum, S., Da, Z. (2019). Indexing and Stock Market Serial Dependence Around the World. *Journal of Financial Economics* 132, 26–48.

// Statement Of Limitations

Not financial advice. This research note is for educational purposes only. It shares an idea and how it was built so that others can study it, work on it and develop something of their own. It is not a finished product and does not constitute an offer or recommendation to trade any instrument. All figures derive from a single MultiCharts backtest export on one NQ contract with a $50,000 nominal balance; commission $5 and slippage $1 per trade. Backtested results are hypothetical, carry the benefit of hindsight and do not represent actual trading. Prop-firm pass rates are simulations from resampled history. Past performance, backtested or realised, does not indicate future results.